

42 Minimally Invasive Lumbar Discectomy

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42.1 Introduction

Minimally invasive techniques have revolutionized the management of pathological conditions in various surgical disciplines and more recently have been applied to the field of spine surgery. One of the most commonly performed procedures is minimally invasive microdiscectomy. Using a muscle-dilating approach, this procedure reduces the iatrogenic soft tissue injury that occurs with routine exposure of the spine. This chapter details the minimally invasive microdiscectomy procedure, including patient selection, operative techniques, complication management, and postoperative care.

42.2 Patient Selection

As with all spinal surgical procedures, proper patient selection begins with a detailed history and physical examination, as well as a judicious review of the radiographic imaging. Once it has been established that the patient has a symptomatic herniated disk, management options should be discussed with the patient. Many patients who have pain from an acutely herniated disk can be satisfactorily treated without surgery. These patients will frequently notice significant improvement after a trial of nonsteroidal anti-inflammatory medication, muscle relaxants, oral steroids, or some combination of these.¹ Physical therapy can also be helpful in ameliorating the pain, particularly in the subacute period. In patients who do not respond to these modalities, epidural steroids can often provide significant pain relief and are commonly given in sets of three. Surgical management should be reserved for patients for whom 6 to 12 weeks of nonoperative therapy have failed or who have a significant motor deficit caused by the disk herniation. The potential risks and complications of the procedure should be explained to the patient, including, but not limited to, nerve root injury, paralysis, back pain, leg pain, failure of symptom relief, neurologic worsening, cerebrospinal fluid (CSF) leakage, spinal instability, reoperation, disk reherniation, infection, spinal instability, and the risks of anesthesia.^{2,3}

42.3 Preoperative Preparation

Minimally invasive microdiscectomy can be performed using epidural anesthesia, but most surgeons prefer to have the patient placed under general anesthesia.⁴ One large-bore intravenous line is usually sufficient, and arterial lines are not normally placed in healthy patients. Intravenous antibiotics should be given before skin incision is done: cefazolin 1 g or vancomycin 1 g in patients who are allergic to penicillin. Compression stockings and thromboembolic disease (TED) hose should be applied prior to turning the patient into the prone position. A Foley catheter is generally not used because the surgery time is usually fairly short. The patient is then carefully turned into the prone position onto a Wilson frame. The frame should be cranked up to encourage flexion of the spine and opening of the interspace. All the pressure points should be appropriately padded, particularly the ulnar regions and the eyes. The anesthesiologist should check the patient's face every 15 minutes for any signs of compression. The lateral C-arm fluoroscope should be brought into the field, preferably from the side of the patient that is opposite the operating microscope. The patient is then prepared and draped in the usual sterile fashion, and the fluoroscope and microscope are also draped at this time.

42.4 Operative Procedure

The universal arm bar is mounted to the bed frame, and the flexible retractor is attached to the arm bar. The location of the surgical level is determined by inserting a 22-gauge spinal needle into the paraspinal soft tissue under fluoroscopic guidance. This is a key step because it will determine the position of the tubular retractor. An 18-mm skin incision created 15 mm lateral to the midline is then centered over this point. The Bovie cautery (Bovie Medical Corporation, St. Petersburg, Florida) is used for hemostasis and to dissect through the dermal layer. A Kirschner wire (K-wire) is passed through the incision and centered on the disk space of interest (► Fig. 42.1). The wire is inserted through the fascia and then the initial dilator is passed over the wire and docked on the lamina-facet junction (► Fig. 42.2 a). The wire should not be used to dock on the spine

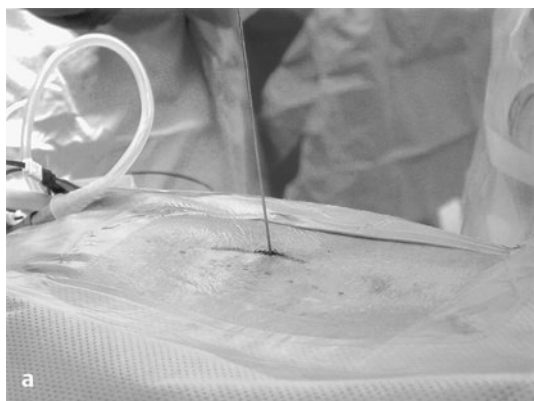


Fig. 42.1 (a) After localizing the skin incision using a 22-gauge spinal needle, the K-wire is inserted through the skin, centered on the disk space. (b) Fluoroscopic image demonstrating the K-wire in place after penetrating the fascia.

because it could inadvertently pass through the interlaminar space and cause a CSF leak or nerve root injury. Once the location of the dilator is confirmed with fluoroscopy, the K-wire is then removed from the field. Controlled forceful sweeping movements of the dilator in the cephalad-caudad and lateral-medial directions are then used to scrape the paraspinal musculature off the bony structures at the tip of the dilator (► Fig. 42.2 b). Incrementally larger dilators are sequentially passed over the initial dilator and advanced to the laminar-facet junction (► Fig. 42.3, ► Fig. 42.4). The numerical reading on the side of the final dilator is used to determine the appropriate tubular retractor length for the patient, ranging from 3 to 10 cm long. The diameter of the tube ranges from 14 to 26 mm, although 16- or 18-mm-diameter retractors are most commonly used for microdiscectomy. The appropriately sized tubular retractor is then passed over the final dilator. A variety of different medical device companies provide slightly different versions of the retractor; essentially, there is no functional difference between the various options. The tubular retractor should be docked on the lamina facet junction such that a small portion of the interlaminar space is visible inferiorly. Once the proper tube position is radiographically confirmed, the tube is locked into position with the flexible retractor arm, and the dilators are removed (► Fig. 42.5). The remainder of the procedure can be performed using loupe magnification, the endoscope, or the operating microscope, and the surgical technique is identical for any of these methods.

The microscope or endoscope is brought into the surgical field at this point and appropriately positioned and focused. The thin layer of soft tissue over the facet and lamina is removed using extended monopolar cautery. A straight or angled curet is used to expose the inferior edge of the lamina, repositioning the tube as needed to obtain proper exposure. A Kerrison punch (KMedic, Northvale, New Jersey) or a drill is then used to perform the laminotomy and medial facetectomy. Leaving the ligamentum flavum intact during the laminotomy decreases the probability of durotomy. Blunt dissection of the ligamentum flavum from the undersurface of the lamina and medial facet is safely and efficiently performed using an angled curet. The pedicle should be palpated (► Fig. 42.6) and serves as a marker for the extent of lateral bone removal. An angled curet or blunt nerve hook is passed rostrally and used to separate the origin of the ligamentum flavum from the underside of the lamina. The ligament is then separated from the dura and gently pulled inferiorly. The Kerrison is then used to remove the ligament in piecemeal fashion. The nerve root is gently retracted medially, exposing the disk space and any free disk fragments. Epidural veins are preserved as much as possible, and electrocautery should be minimized to reduce the amount of postoperative epidural scarring. Injectable hemostatic agents generally provide excellent hemostasis. The microknife is used to incise the annulus, and the disk herniation is removed using a combination of a small pituitary forcep and a sucker using a hand-over-fist technique. Ideally, the disk herniation should be gently

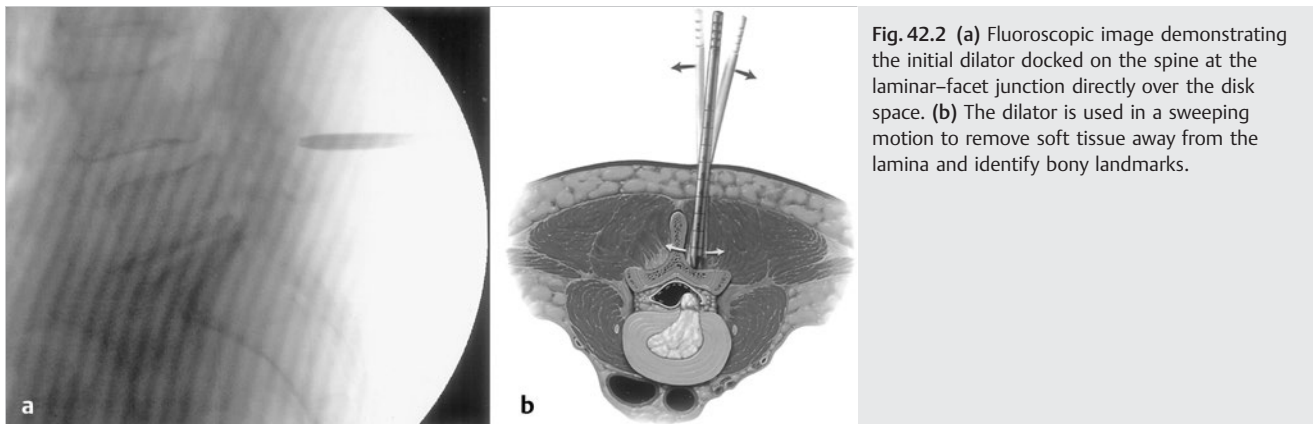


Fig. 42.2 (a) Fluoroscopic image demonstrating the initial dilator docked on the spine at the laminar-facet junction directly over the disk space. (b) The dilator is used in a sweeping motion to remove soft tissue away from the lamina and identify bony landmarks.

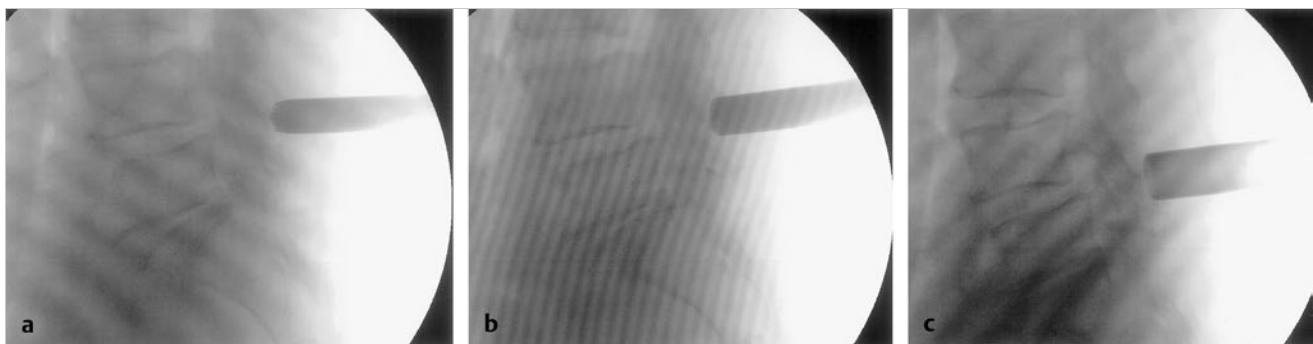


Fig. 42.3 Serial fluoroscopic images demonstrating gradual dilation of muscle and fascia in preparation of tubular retractor placement. (a) A 14-mm diameter dilator is centered over the L4-5 interspace. (b) A 16-mm dilator has now been introduced into the wound. (c) The final dilator has been placed, and next the tubular retractor will be inserted.

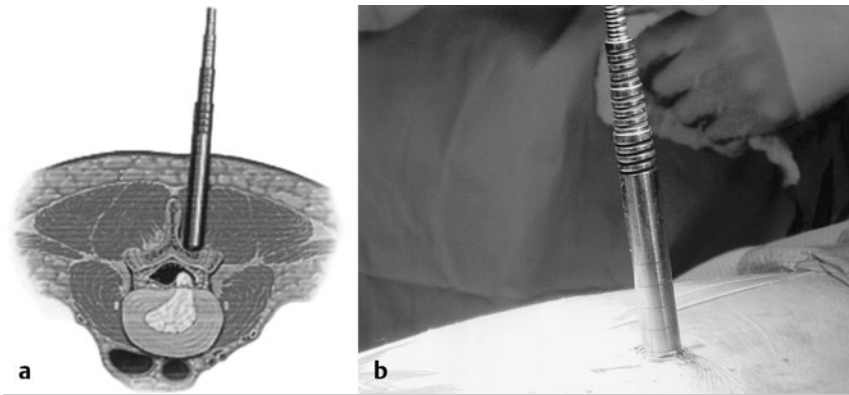


Fig. 42.4 (a) Schematic and (b) operating room photograph showing serial dilators in proper position.

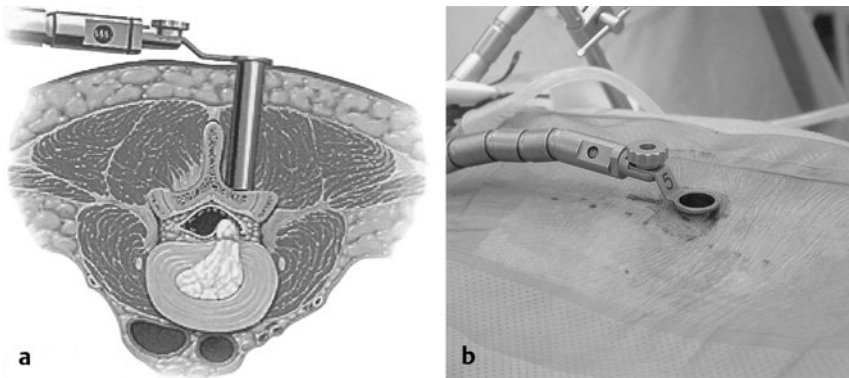


Fig. 42.5 (a) Schematic and (b) operating room photograph demonstrating a 5-cm-length tubular retractor being held in position by a flexible retractor arm that is connected to a bed-mounted universal bar.

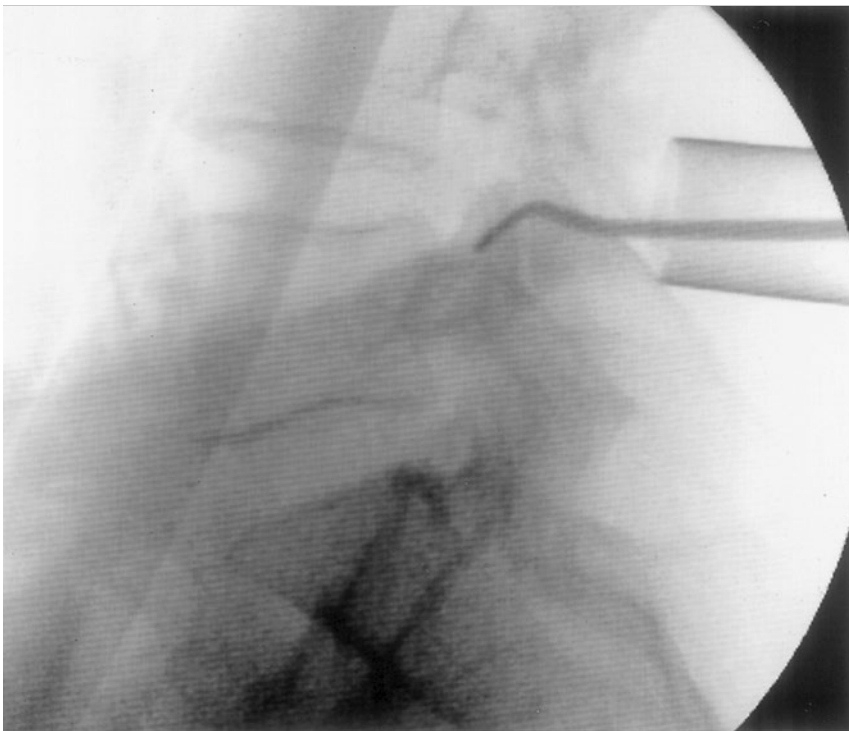


Fig. 42.6 A Woodson instrument is used to palpate the pedicle, which serves as a landmark for the lateral extent of bone removal.

removed in one piece without tearing it apart and to avoid leaving behind residual disk fragments that may retract underneath the thecal sac. Curets may be used to help facilitate removal of firm or calcified disk herniations. Once the disk herniation has been satisfactorily removed, a Woodson instrument or ball

hook may be used to explore the canal and the foramen for any remaining evidence of compression.

The wound is then copiously irrigated with antibiotic saline, and careful hemostasis is once again achieved. The flexible retractor arm is then loosened and the tube is gently

withdrawn under direct vision, taking care to cauterize any bleeding spots within the soft tissue. The fascia is then closed with one or two interrupted absorbable 0 Vicryl sutures (Ethicon, Inc., Somerville, New Jersey), followed by closure of the dermal layer with 3–0 Vicryl and a subcuticular layer with 4–0 Monocryl (Ethicon, Inc.). An intermediate acting local anesthetic can be injected into the soft tissue. The skin is then dressed with Steri-Strips (3 M, St. Paul, Minnesota) and a sterile bandage.

42.5 Postoperative Management

After full emergence from general anesthesia a thorough neurologic examination is performed in the recovery room and documented. The patient should be on bed rest for the first 2 hours postoperatively and then may get out of bed and ambulate. Discharge can occur 3 hours postoperatively after the patient is able to ambulate, void, and tolerate a liquid diet. Discharge medications include a stool softener, a muscle relaxant, and a mild narcotic pain reliever. Nonsteroidal anti-inflammatory medication can also be given at the discretion of the surgeon. The patients are encouraged to walk for exercise the following day but are instructed to avoid bending, twisting, or lifting anything heavier than 10 pounds for about 1 to 2 weeks postoperatively, followed by gradual resumption of normal activities over the next 4 to 6 weeks.

The complication rate for minimally invasive microdiscectomy is certainly higher in the early stages of a surgeon's learning curve but should be equal to that of traditional microdiscectomy after adequate experience has been obtained.³

Although technically difficult, dural tears can be primarily sewn through the tube if located dorsally.⁵ Small dural tears can also be treated with collagen sponge, fibrin glue, and 24 to 48 hours of bed rest. In cases of large dural tears, the incision may have to be formally opened to adequately treat the CSF leak. Other complications, such as infection, neurologic deficit, and disk reherniation, should be managed in the same manner as traditionally performed microdiscectomy.

42.6 Conclusion

Minimally invasive discectomy offers the benefit of less soft tissue disruption and, in experienced hands, similar efficacy, compared with the open technique.

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